Simple CMOS Process

Note: In this document for flow, all of the Photolithography steps should be judged by yourself. If the Photolithography steps are not necessary in the simulation, you can not write down the codes for photolithography.
If there are extra unnecessary Photolithography codes in your report, it will be penalized in scores.

28.	POST oxidation (10±4nm) 3-1# 2260SPAC	3# Clean Oxidation into furnace Oxidation outof furnace	If you have error in this step, check mesh	Thickness of oxide on top of NMOS Poly and PMOS Poly
29.	N+S/D Photolithography Mask：NPLUS	Pretreatment Photo deposit Exposure Development Check		
30.	Thermosetting membrane		（No simulation）	
31.	N+ S/D implant		As 80Kev 5 E15 It's suggest that x.val for N+ xj is far away from bird beak.	Junction depth of N+
32.	Remove all of photo	Dry etch photo Wet etch photo check		
33.	POST oxidation (10±4nm) 3-1# 2260SPAC	3# Clean Oxidation into furnace Oxidation outof furnace		Thickness of oxide on top of NMOS Poly and PMOS Poly
34.	P+S/D Photolithography Mask：PPLUS	Pretreatment Photo deposit Exposure Development Check		
35.	Thermosetting membrane		（No simulation）	
36.	P+ S/D implant		BF_2 (the impurity is written as bf2) 80Kev 4E15 It's suggest that x.val for P+ xj is far away from bird beak.	Junction depth of P+
37.	Remove all of photo	Dry etch photo Wet etch photo check		
38.	PMD deposit	3# Clean USG 400±20nm BPSG 600±60nm	In simulation, 1000nm SiO2 is deposited	
39.	BPSG	Oxidation into furnace	（No simulation）	

CMOS 工 艺 流 程 单

批号： 电路名称： 拟制：

	4-2# BPSGDENS	Oxidation outof furnace						
40.	CONT Photolithography Mask：CONT	Pretreatment	Cont is a hole in the oxide layer that allows the metal to contact with the silicon substrate					
		Photo deposit						
		Exposure						
		Development						
		Check						
41.	CONT etch	UV hard bake	（No simulation）					
		Cont etch	Etch SiO2					
42.	Remove all of photo	Dry etch photo						
		Wet etch photo						
		check						
43.	Reflow 4-2# BPSGFLOW	3# Clean	（No simulation）					
		Oxidation into furnace						
		Oxidation outof furnace						
44.	Deposit Al	BOE 28 seconds	（No simulation）					
		Metal sputter	1450±110nm					
45.	Al Photolithography Mask：Metal	Pretreatment						
		Photo deposit						
		Exposure						
		Development						
		Check						
46.	Al etch	UV hard bake						
		Metal etch						
		clean						
47.	Remove all of photo	Dry etch photo						As follows
		Wet etch photo						
		check						
1. Vt for NMOS with Vsub=2V； 2. Vt for PMOS with Vsub=0V; 3. Channel doping for NMOS and PMOS; 4. Gate oxide thickness for NMOS and PMOS								
No simulation from this step								
48.	Passivation deposit SiO2(550±50nm) SiN(300±30nm)	DI Clean						
		SiO_2 deposit						
		Si_3N_4 deposit						
49.	Passivation Photolithography	Pretreatment						
		Photo deposit						
		Exposure						
		Development						

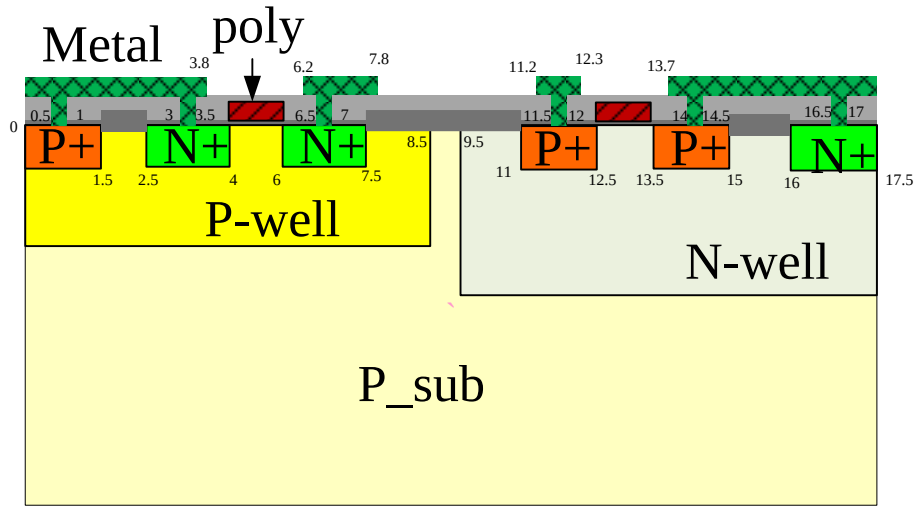
CMOS 工 艺 流 程 单

批号： 电路名称： 拟制：

50.	Passivation etch SiO2(550±50nm) SiN(300±30nm)	hard bake						
		Si_3N_4 etch						
		SiO_2 etch						
		Dry etch photo						
		EKC Clean						
51.	420℃ ally 4-4# 420ALLY	Check Al						
		DI Clean						
		Oxidation into furnace						
		Oxidation outof furnace						

2260SPAC 3-1 100±40A post oxidation								
Step	Temp °C	Ramp °C/min	Time	N2(slpm)	O2(slpm)	H2(slpm)	Hcl (sccm)	Boat Cm/min
Push in	800		18’	8				
Sta temp	800		15’	8				
Ramp to	900		15’	8				
Hcl dryox	900		15’30”		8		100	
Hcl off	900		3’		8			
Ramp to	800		30’	8				
Push out	800		15’	15				

Note: 100 sccm =0.1slpm



NWELL PWELL ACTIVE NPLUS PPLUS POLY CONT METAL